EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 5

1. **Voter paradox :** Let X, Y, Z be discrete random variables with the property that their values are distinct with probability 1. Let $a = \mathbb{P}(X > Y), b = \mathbb{P}(Y > Z), c = \mathbb{P}(Z > X)$.

   1. Show that $\min\{a, b, c\} \leq 2/3$. Give an example where the equality is attained.
   2. Can X, Y, Z be independent and identically distributed and still be distinct with probability 1?

Solution:

1. Let $I(k)$ be the indicator random variable which is defined as:

$$\begin{cases} 1 & \text{if statement } k \text{ is true} \\ 0 & \text{if statement } k \text{ is false} \end{cases}$$

   Using this definition, we can write:

$$\begin{aligned} a + b + c &= P(X > Y) + P(Y > Z) + P(Z > X) \\ &= E[I(X > Y) + I(Y > Z) + I(Z > X)] \end{aligned}$$

   It is also trivial to see that $I(X > Y) + I(Y > Z) + I(Z > X)$ can be at most 2 since at most any two of the statements can be true. We have:

$$I(X > Y) + I(Y > Z) + I(Z > X) \leq 2$$

$$a + b + c = E[I(X > Y) + I(Y > Z) + I(Z > X)] \leq 2$$

   Using the property of minimum and mean, we have:

$$\min\{a, b, c\} \leq \frac{a + b + c}{3} \leq \frac{2}{3}$$

   Example: Let $\mathbb{P}(X = 3, Y = 2, Z = 1) = \mathbb{P}(X = 1, Y = 3, Z = 2) = \mathbb{P}(X = 2, Y = 1, Z = 3) = \frac{1}{3}$. Clearly, we have $a = b = c = \frac{2}{3}$ and hence, $\min\{a, b, c\} = \frac{2}{3}$.

2. Let $f(x)$ denote the marginal PMF of the IID RVs X, Y, Z. Since they are IID RVs, we have $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X)\mathbb{P}(Y)$. It is also given that the random variables are distinct with probability 1, i.e., $\mathbb{P}(X = Y) = 0$. Since X, Y are discrete RVs, this can be written as:

$$\mathbb{P}(X = Y) = \sum_{j \in D} \mathbb{P}(X = j, Y = j) = 0$$

   where D is the domain of the RVs X, Y, Z. We have:

$$\sum_{j \in D} \mathbb{P}(X = j, Y = j) = \sum_{j \in D} \mathbb{P}(X = j)\mathbb{P}(Y = j) = \sum_{j \in D} (f(j))^2 = 0$$

   This can be true only if $f(j) = 0$ for all $j \in D$. However, this is clearly a contradiction since $\sum_{j \in D} f(j) = 1$. Hence, discrete IID RVs cannot satisfy the property that they are distinct with probability 1.

2. Let X have a Pareto distribution with scale parameter $x_m = 1$ and shape parameter $\alpha = 2$. Its probability density function is

$$f_X(x) = \frac{2}{x^3}, \qquad \infty > x \geq 1.$$

Answer the following:

1. Show that $f_X(x)$ is a valid probability density function by proving that it integrates to 1.
2. Compute $\mathbb{E}[X]$.
3. Compute the second moment $\mathbb{E}[X^2]$.

Solution:

1. Normalization:

$$\int_1^\infty f_X(x)\,dx = \int_1^\infty \frac{2}{x^3}\,dx = 2\left[-\frac{1}{2x^2}\right]_1^\infty = (0-(-1)) = 1.$$

Hence, $f_X(x)$ integrates to 1 and is a valid PDF.

2. Mean:

$$\mathbb{E}[X] = \int_1^\infty x \cdot \frac{2}{x^3}\,dx = 2\int_1^\infty x^{-2}\,dx = 2\left[-\frac{1}{x}\right]_1^\infty = 2(0-(-1)) = 2.$$

Thus, $\mathbb{E}[X] = 2$ (finite).

3. Second Moment:

$$\mathbb{E}[X^2] = \int_1^\infty x^2 \cdot \frac{2}{x^3}\,dx = 2\int_1^\infty \frac{1}{x}\,dx = 2\lim_{b\to\infty}\int_1^b \frac{1}{x}\,dx = 2\lim_{b\to\infty}(\ln b - \ln 1).$$

Since $\ln b \to \infty$ as $b \to \infty$, we conclude

$$\mathbb{E}[X^2] = \infty.$$

3. Let X be a nonnegative continuous random variable with cumulative distribution function $F(x) = \Pr(X \leq x)$. Prove that

$$\mathbb{E}[X] = \int_0^\infty \big(1 - F(x)\big)\,dx.$$

Solution: We start from the right-hand side:

$$\int_0^\infty \big(1 - F(x)\big)\,dx = \int_0^\infty \Pr(X > x)\,dx = \int_x^\infty f(t)\,dt$$

Swapping the order of integration:

$$\int_0^\infty \left(\int_x^\infty f(t)\,dt\right) dx = \int_0^\infty \left(\int_0^t dx\right) f(t)\,dt.$$

$$= \int_0^\infty t f(t)\,dt = \mathbb{E}[X]$$

4. Consider a sequence of independent coin flips, each of which has probability p of being heads. Define a random variable X as the length of the run (of either heads or tails) started by the first trial. (For example, $X = 3$ if either TTTH or HHHT is observed.) Find the distribution of X, and find $E[X]$.

Solution:

(a) The event $X = k$ means the first flip is one outcome, the next $k-1$ flips match the first, and the k-th flip is the opposite. We use the law of total probability, conditioning on the first flip:

$$P(X = k) = P(X = k \mid \text{1st is H})P(\text{1st is H}) + P(X = k \mid \text{1st is T})P(\text{1st is T})$$

1. If the first flip is H (probability p): For $X = k$, the sequence is $\underbrace{\text{H, H, } \ldots \text{, H}}_{k-1}$ T.

$$P(X = k \mid \text{1st is H}) = p^{k-1}q$$

2. If the first flip is T (probability q): For $X = k$, the sequence is $\underbrace{\text{T, T, } \ldots \text{, T}}_{k-1}$ H.

$$P(X = k \mid \text{1st is T}) = q^{k-1}p$$

Substituting these back for $k = 1, 2, 3, \ldots$:

$$\begin{aligned} P(X = k) &= (p^{k-1}q)p + (q^{k-1}p)q \\ &= p^k q + q^k p \\ &= pq(p^{k-1} + q^{k-1}) \end{aligned}$$

$\therefore$ The distribution of X is:

$$P(X = k) = pq(p^{k-1} + q^{k-1}), \quad k \in \{1, 2, 3, \ldots\}$$

(b) The expected value, $\mathbb{E}[X] = \sum_{k=1}^{\infty} kP(X = k)$:

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k(p^k q + q^k p) = pq \sum_{k=1}^{\infty} kp^{k-1} + pq \sum_{k=1}^{\infty} kq^{k-1}$$

We use the identity for the derivative of the geometric series: $\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$.

- $\sum_{k=1}^{\infty} kp^{k-1} = \frac{1}{(1-p)^2} = \frac{1}{q^2}$
- $\sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2} = \frac{1}{p^2}$

Substituting these results:

$$\begin{aligned} \mathbb{E}[X] &= pq\left(\frac{1}{q^2} + \frac{1}{p^2}\right) \\ &= pq\left(\frac{p^2 + q^2}{p^2 q^2}\right) \\ &= \frac{p^2 + q^2}{pq} = \frac{2p^2 - 2p + 1}{p - p^2} \end{aligned}$$

5. Prove that if the kth absolute moment of a random variable X is not finite (i.e., $\mathbb{E}[|X|^k] = \infty$ for some

$k > 0$), then for every $m > k$, the mth absolute moment is also not finite: $\mathbb{E}[|X|^m] = \infty$.

Solution: Fix $k > 0$ and let $m > k$. Split the space according to $\{|X| \leq 1\}$ and $\{|X| > 1\}$. Note that on $\{|X| > 1\}$ we have $|X|^m \geq |X|^k$, while on $\{|X| \leq 1\}$ we have $|X|^k \leq 1$, hence

$$\mathbb{E}\big[|X|^k \, \mathbf{1}_{\{|X|\leq 1\}}\big] \leq \mathbb{E}[\mathbf{1}_{\{|X|\leq 1\}}] \leq 1 < \infty.$$

Therefore, if $\mathbb{E}[|X|^k] = \infty$, it must be that

$$\mathbb{E}\big[|X|^k \, \mathbf{1}_{\{|X|>1\}}\big] = \infty.$$

Using the monotonicity on $\{|X| > 1\}$,

$$\mathbb{E}[|X|^m] \;\geq\; \mathbb{E}\big[|X|^m \, \mathbf{1}_{\{|X|>1\}}\big] \;\geq\; \mathbb{E}\big[|X|^k \, \mathbf{1}_{\{|X|>1\}}\big] = \infty.$$

Hence $\mathbb{E}[|X|^m] = \infty$ for every $m > k$.

6. Four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively, 40, 33, 25, and 50 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. Let Y denote the number of students on a randomly selected bus. Which of $\mathbb{E}[X]$ or $\mathbb{E}[Y]$ is larger?

Solution: Let X be a random variable for the number of students in the randomly selected student-traveling bus. X can take the values $40, 33, 25, 50$. Since picking any student is equally likely, we have an expected value of X as follows:

$$\begin{aligned}\mathbb{E}[X] &= 40\mathbb{P}[X = 40] + 33\mathbb{P}[X = 33] + 25\mathbb{P}[X = 25] + 50\mathbb{P}[X = 50]. \\ &= 40\left(\frac{40}{148}\right) + 33\left(\frac{33}{148}\right) + 25\left(\frac{25}{148}\right) + 50\left(\frac{50}{148}\right).\end{aligned}$$

Let Y be a random variable of number of students in the randomly selected bus driver. Y also can take the values $40, 33, 25, 50$. Since picking any driver is equally likely, we have expected values of Y as follows:

$$\begin{aligned}\mathbb{E}[Y] &= 40\mathbb{P}[Y = 40] + 33\mathbb{P}[Y = 33] + 25\mathbb{P}[Y = 25] + 50\mathbb{P}[Y = 50]. \\ &= 40\left(\frac{1}{4}\right) + 33\left(\frac{1}{4}\right) + 25\left(\frac{1}{4}\right) + 50\left(\frac{1}{4}\right).\end{aligned}$$

Clearly, $\mathbb{E}[X] > \mathbb{E}[Y]$. This is because, the students selected are more likely to come from the bus carrying a maximum number of students, where the drivers are equally likely to be selected.

7. Two friends A and B plan to meet at a park. Both of them reach the park between 5:00 pm and 6:00 pm with a uniform distribution and independent of each other. Find the distribution of the waiting time of A and the expected value of the waiting time of A. (**Hint**: The waiting time of A is zero if B is already in the park when A reaches the park.)

Solution:

- **Distribution of the Waiting Time of A** Let X and Y represent the arrival times of A and B, respectively, where both $X, Y \in [0, 60]$ minutes. The arrival times are independently and uniformly distributed on the interval $[0, 60]$. Thus, the joint distribution is:

$$f_{X,Y}(x, y) = \frac{1}{3600}, \quad \text{for } 0 \leq x, y \leq 60.$$

  The waiting time of A, denoted W_A, is the amount of time A has to wait for B after arriving:

$$W_A = \max(0, Y - X).$$

  This means A waits for $Y - X$ minutes if B arrives later, and the waiting time is zero if B arrives earlier or at the same time.

- **Expected Value of the Waiting Time of A** Since X and Y are independent, the probability that A arrives before B is $\frac{1}{2}$. We need to calculate the expected waiting time $E[W_A]$.

$$\begin{aligned} E[W_A] &= \int_0^{60} \int_x^{60} (y - x) \cdot \frac{1}{3600} \, dy \, dx \\ &= \frac{1}{3600} \int_0^{60} \int_x^{60} (y - x) \, dy \, dx. \end{aligned}$$

  Evaluating the inner integral:

$$\int_x^{60} (y - x) \, dy = \left[\frac{(y - x)^2}{2} \right]_x^{60} = \frac{(60 - x)^2}{2}.$$

  Now, substitute this into the outer integral:

$$E[W_A] = \frac{1}{7200} \int_0^{60} (60 - x)^2 \, dx.$$

  Evaluating the outer integral:

$$\begin{aligned} E[W_A] &= \frac{1}{7200} \int_0^{60} (60 - x)^2 \, dx \\ &= \frac{1}{7200} \left[\frac{(60 - x)^3}{3} \right]_0^{60} \\ &= \frac{1}{7200} \left(0 - \frac{60^3}{3} \right) \\ &= \frac{1}{7200} \cdot \left(-\frac{216000}{3} \right) \\ &= \frac{216000}{21600} = 10. \end{aligned}$$

- **Result** The expected value of the waiting time of A is:

$$E[W_A] = 10 \text{ minutes}.$$

8. Aravind is a market-maker who must decide what bid (buy price) b and ask (sell price) a to quote for an asset. The true value of the asset, denoted V, is chosen uniformly at random by Haricharan in $[0, 1]$.

- 500 uninformed traders: each draws $U \sim \text{Unif}[0,1]$. If $U > a$, the uninformed trader buys at Aravind's ask a. If $U < b$, the uninformed trader sells at Aravind's bid b. Otherwise, no trade occurs.
- 500 informed traders: each knows V exactly and will buy if $V > a$, sell if $V < b$, and not trade otherwise.

Find the optimal bid b and ask a that Aravind should quote in order to maximize his expected profit when trading with all 1000 traders.

Solution: The true (fair) value V is a random variable with distribution

$$V \sim \text{Unif}[0,1].$$

So Aravind should post a bid- that is symmetric around the midpoint

$$m = \mathbb{E}[V] = \tfrac{1}{2}.$$

Let the bid–ask spread (difference between sell and buy price) be x, with quotes $b = 0.5 - \frac{x}{2}$ and $a = 0.5 + \frac{x}{2}$.

- An uninformed trader trades with probability $1 - x$, yielding an expected profit of $\frac{x}{2}$ per trade; hence their Profit from the profitable trades is $\Pi_U = 500(1-x)\frac{x}{2}$.
- An informed trader also trades with probability $1 - x$, but the market maker then loses $\frac{1-a}{2}$ when $V > a$ and $\frac{b}{2}$ when $V < b$. Substituting $a = 0.5 + \frac{x}{2}$ and $b = 0.5 - \frac{x}{2}$ gives an expected loss $\frac{1}{4} - \frac{x}{4}$, so $\Pi_I = 500(1-x)\big(\frac{1}{4} - \frac{x}{4}\big)$.

The net profit is therefore

$$\Pi(x) = \Pi_U - \Pi_I = 500(1-x)\Big(\tfrac{3x}{4} - \tfrac{1}{4}\Big).$$

Maximizing, we find $g'(x) = -\frac{3}{2}x + 1 = 0$, so the optimal spread is $x^* = \frac{2}{3}$, with the corresponding quotes $b = \frac{1}{6}$ and $a = \frac{5}{6}$.

$$\boxed{\text{Optimal bid–ask spread: } x^* = 2/3, \;\; b = 1/6, \;\; a = 5/6.}$$

9. Consider a rectangle of length T (along the x-axis) and width d (along the y-axis),

$$R = \{(x,y) : 0 \le x \le T,\ 0 \le y \le d\}.$$

The two horizontal sides $y = 0$ and $y = d$ play the role of the nearest parallel lines. A needle of length ℓ (with $\ell \le d$) is dropped uniformly at random into R with a uniformly random orientation (meaning the center of the needle always lies in R). Let X be the indicator of the event that the needle intersects either of the horizontal sides. Compute $\Pr(X = 1)$.

Solution: Let the needle's center have coordinates (X_c, Y_c), chosen uniformly in R. Let Θ denote the acute angle between the needle and the horizontal direction; by symmetry we may take Θ uniform on $[0, \pi/2]$. The joint density of (Y_c, Θ) (marginalizing X_c which does not affect crossing) is

$$f_{Y_c,\Theta}(y,\theta) = \frac{2}{d} \cdot \frac{2}{\pi} = \frac{4}{\pi d}, \qquad 0 \le y \le \frac{d}{2},\ 0 \le \theta \le \frac{\pi}{2},$$

since we may restrict Y_c to the distance to the nearer horizontal side, which lies in $[0, d/2]$, and reflect symmetry vertically.

The needle crosses a horizontal side if and only if the vertical projection of half the needle exceeds the centre's distance to the nearer side. Equivalently,

$$y \le \frac{\ell}{2}\sin\theta.$$

Therefore, for $\ell \le d$,

$$\begin{aligned}
\Pr(\text{needle crosses}) &= \iint_{\{(y,\theta):\, 0\le y\le \frac{\ell}{2}\sin\theta,\ 0\le\theta\le\pi/2\}} \frac{4}{\pi d}\, dy\, d\theta \\
&= \frac{4}{\pi d}\int_0^{\pi/2}\int_0^{(\ell/2)\sin\theta} dy\, d\theta \\
&= \frac{4}{\pi d}\int_0^{\pi/2} \frac{\ell}{2}\sin\theta\, d\theta \\
&= \frac{2\ell}{\pi d}\int_0^{\pi/2} \sin\theta\, d\theta = \frac{2\ell}{\pi d}\cdot 1 = \frac{2\ell}{\pi d}.
\end{aligned}$$

$$\boxed{\Pr(\text{needle crosses a horizontal side}) = \frac{2\ell}{\pi d} \quad (\ell \le d)}$$

10. Suppose that $X_1, X_2, \ldots, X_n$ are random variables such that the variance of each variable is 1 and the correlation between each pair of different variables is $\frac{1}{4}$. Determine $\mathrm{Var}(X_1 + X_2 + \cdots + X_n)$.

Solution: We are given

$$\mathrm{Var}(X_i) = 1, \quad \text{for all } i,$$

and for $i \ne j$,

$$\mathrm{Corr}(X_i, X_j) = \tfrac{1}{4}.$$

Thus,

$$\mathrm{Cov}(X_i, X_j) = \tfrac{1}{4}\sqrt{1\cdot 1} = \tfrac{1}{4}.$$

Now,

$$\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathrm{Var}(X_i) \;+\; 2\sum_{1\le i<j\le n} \mathrm{Cov}(X_i, X_j).$$

The first sum gives

$$\sum_{i=1}^n \mathrm{Var}(X_i) = n.$$

The second term is

$$2\cdot\binom{n}{2}\cdot\frac{1}{4} = 2\cdot\frac{n(n-1)}{2}\cdot\frac{1}{4} = \frac{n(n-1)}{4}.$$

Therefore,

$$\mathrm{Var}(X_1 + X_2 + \cdots + X_n) = n + \frac{n(n-1)}{4} = \frac{n^2+3n}{4}.$$

$$\boxed{\text{Var}(X_1 + \cdots + X_n) = \frac{n(n+3)}{4}}$$

11. Let $X \sim$ Bernoulli(0.4), $Y \sim$ Bernoulli(0.6) be independent. Suppose $Z = (X + Y)$ and $W = |X - Y|$. Determine the following:

    1. $\text{Var}(Z)$ and $\text{Var}(W)$
    2. $\text{Cov}(Z, W)$
    3. $\rho(Z, W)$

Solution: Clearly, $\Omega_X = \{0, 1\}$ with $p_X(0) = 0.6,\ p_X(1) = 0.4$, and $\Omega_Y = \{0, 1\}$ with $p_Y(0) = 0.4,\ p_Y(1) = 0.6$.

1. The random variables are

$$Z = X + Y \in \{0, 1, 2\}, \quad W = |X - Y| \in \{0, 1\}.$$

The possible ordered pairs (X, Y), their probabilities, and corresponding values of Z and W are given below:

X	Y	$p_{XY}(x, y)$	$Z = X + Y$	$W = \lvert X - Y \rvert$
0	0	0.24	0	0
0	1	0.36	1	1
1	0	0.16	1	1
1	1	0.24	2	0

Since X and Y are independent, $p_{XY}(x, y) = p_X(x)p_Y(y)$.

The joint support of (Z, W) is $\Omega_{ZW} = \{(0, 0), (1, 1), (2, 0)\}$ with joint PMF:

ZW	0	1
0	0.24	0
1	0	0.52
2	0.24	0

The marginals are:

$$p_Z(0) = 0.24,\ p_Z(1) = 0.52,\ p_Z(2) = 0.24, \qquad p_W(0) = 0.48,\ p_W(1) = 0.52.$$

For Z:

$$\mu_Z = \mathbb{E}[Z] = 1, \quad \mathbb{E}[Z^2] = 1.48,$$
$$\text{Var}(Z) = 1.48 - 1^2 = 0.48.$$

For W:

$$\mu_W = \mathbb{E}[W] = 0.52, \quad \mathbb{E}[W^2] = 0.52,$$
$$\text{Var}(W) = 0.52 - (0.52)^2 = 0.2496.$$

2. Compute $\mathbb{E}[ZW]$:

$$\mathbb{E}[ZW] = \sum_{z,w} (zw)\, p_{ZW}(z,w) = 1 \cdot 1 \cdot 0.52 = 0.52.$$

Thus

$$\mathrm{Cov}(Z,W) = \mathbb{E}[ZW] - \mu_Z \mu_W = 0.52 - (1 \times 0.52) = 0.$$

3. Finally,

$$\rho(Z,W) = \frac{\mathrm{Cov}(Z,W)}{\sqrt{\mathrm{Var}(Z)}\sqrt{\mathrm{Var}(W)}} = \frac{0}{\sqrt{0.48}\sqrt{0.2496}} = 0.$$

12. Consider a group of n roommate pairs at a college (so there are $2n$ students). Each of these $2n$ students independently decides randomly whether to take a certain course, with probability p of success (where "success" is defined as taking the course). Let N be the number of students among these $2n$ who take the course, and let X be the number of roommate pairs where both roommates in the pair take the course. Find $E(X)$ and $E(X \mid N)$.

Solution: Create an indicator r.v. I_j for the jth roommate pair, equal to 1 if both take the course. The expected value of such an indicator r.v. is p^2, so $E(X) = np^2$ by symmetry and linearity. Similarly, $E(X \mid N) = nE(I_1 \mid N)$. We have

$$E(I_1 \mid N) = \frac{N}{2n} \cdot \frac{N-1}{2n-1}$$

since given that N of the $2n$ students take the course, the probability is $\frac{N}{2n}$ that any particular student takes the course(the p no longer matters), and given that one particular student in a roommate pair takes the course, the probability that the other roommate does is $\frac{N-1}{2n-1}$. Thus,

$$E(X \mid N) = nE(I_1 \mid N) = \frac{N(N-1)}{2} \cdot \frac{1}{2n-1}.$$

13. You find yourself in a room with three exits in the centre of a maze.

- Exit 1 leads outside the maze after 2 minutes.
- Exit 2 leads back to the room after 4 minutes.
- Exit 3 leads back to the room after 6 minutes.

Assume that every time you make a choice, you're equally likely to choose any of the three exits. What is the expected time taken by you to leave the maze?

Solution: Let X be the time taken by you to leave the maze starting from the room, and Y be the exit you choose first. Y takes the values 1, 2 and 3. Then we have:

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}_Y[\mathbb{E}[X|Y]] \\
&= \sum_{y=1}^{3} \mathbb{E}[X|Y=y]\mathbb{P}(Y=y) \\
&= \mathbb{E}[X|Y=1] \cdot \frac{1}{3} + \mathbb{E}[X|Y=2] \cdot \frac{1}{3} + \mathbb{E}[X|Y=3] \cdot \frac{1}{3}
\end{aligned}$$

Notice (all in minutes):

- $\mathbb{E}[X|Y=1] = 2$
- $\mathbb{E}[X|Y=2] = 4 + \mathbb{E}[X]$
- $\mathbb{E}[X|Y=3] = 6 + \mathbb{E}[X]$

Plugging in above expressions for the conditional expectation and solving for $\mathbb{E}[X]$ gives $\mathbb{E}[X] =$ 12 minutes.

14. 100 exam papers are on the table, with two questions to be graded. Sparsh is in charge of grading question one and Deepak is in charge of grading question two. First, Sparsh grades some exam papers at random; each exam paper has probability 0.3 of being graded. Next, Deepak randomly grades half of the exam papers. That is, he grades 50 exam papers. Assume Sparsh and Deepak make their choices independently.
    (a) Let N be the number of exam papers that Sparsh graded. What is $\mathbb{E}[N]$?
    (b) Let M be the number of exam papers that Sparsh graded and Deepak did not grade. What is $\mathbb{E}[M]$?

Solution:

(a) Let N_i be the indicator random variable for the event that Sparsh grades the ith exam paper:

$$N_i = \begin{cases} 1 & \text{if Sparsh grades the } i\text{th paper,} \\ 0 & \text{otherwise.} \end{cases}$$

Then $\mathbb{E}[N_i] = 0.3$. Since $N = \sum_{i=1}^{100} N_i$, we get

$$\mathbb{E}[N] = \mathbb{E}\left[\sum_{i=1}^{100} N_i\right] = \sum_{i=1}^{100} \mathbb{E}[N_i] = 100 \cdot 0.3 = 30.$$

(b) Let M_i be the indicator random variable for the event that the ith exam paper is graded by Sparsh and not by Deepak:

$$M_i = \begin{cases} 1 & \text{if the } i\text{th paper is graded by Sparsh but not by Deepak,} \\ 0 & \text{otherwise.} \end{cases}$$

The probability of this event is

$$\Pr(M_i = 1) = 0.3 \cdot (1 - 0.5) = 0.15,$$

so $\mathbb{E}[M_i] = 0.15$. Since $M = \sum_{i=1}^{100} M_i$, we get

$$\mathbb{E}[M] = \mathbb{E}\left[\sum_{i=1}^{100} M_i\right] = \sum_{i=1}^{100} \mathbb{E}[M_i] = 100 \cdot 0.15 = 15.$$

15. (a) Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with the uniform distribution on $[0, 1]$.

1. Give the cumulative distribution function, the probability density function, and the expected value of
$$Z_n = \max(X_1, \ldots, X_n).$$
2. Give the cumulative distribution function, the probability density function, and the expected value of
$$Y_n = \min(X_1, \ldots, X_n).$$

(b) 500 ants are independently placed uniformly at random on a 1-foot string. Each ant independently chooses a direction (left or right) with probability 1/2 and then moves at speed 1 foot/minute until it falls off an end. When two ants meet head-on they instantly reverse directions and continue at speed 1. What is the expected time until all ants have fallen off the string? (Hint, use the results obtained in part a)

Solution:

(a) 1. **Maximum Z_n.**
For $0 \le z \le 1$,
$$F_{Z_n}(z) = \Pr(Z_n \le z) = \Pr(X_1 \le z, \ldots, X_n \le z) = \Pr(X_1 \le z) \cdots \Pr(X_n \le z) = z^n.$$
Hence the PDF is
$$f_{Z_n}(z) = \frac{d}{dz} F_{Z_n}(z) = n z^{n-1}, \qquad 0 \le z \le 1.$$
The expectation is
$$\mathbb{E}[Z_n] = \int_0^1 z \cdot n z^{n-1}\, dz = n \int_0^1 z^n\, dz = n \cdot \frac{1}{n+1} = \frac{n}{n+1}.$$
$$\boxed{F_{Z_n}(z) = z^n, \quad f_{Z_n}(z) = n z^{n-1}, \quad \mathbb{E}[Z_n] = \frac{n}{n+1}.}$$

2. **Minimum Y_n.**
For $0 \le y \le 1$,
$$F_{Y_n}(y) = \Pr(Y_n \le y) = 1 - \Pr(Y_n > y) = 1 - \Pr(X_1 > y, \ldots, X_n > y).$$
Since $\Pr(X_i > y) = 1 - y$,
$$F_{Y_n}(y) = 1 - (1-y)^n.$$
Differentiate to get the PDF:
$$f_{Y_n}(y) = \frac{d}{dy} F_{Y_n}(y) = n(1-y)^{n-1}, \qquad 0 \le y \le 1.$$
The expectation is
$$\mathbb{E}[Y_n] = \int_0^1 y \cdot n(1-y)^{n-1}\, dy.$$
Use the Beta integral (or symmetry) to obtain
$$\mathbb{E}[Y_n] = \frac{1}{n+1}.$$
$$\boxed{F_{Y_n}(y) = 1 - (1-y)^n, \quad f_{Y_n}(y) = n(1-y)^{n-1}, \quad \mathbb{E}[Y_n] = \frac{1}{n+1}.}$$

(b) **Ants on a string.** First note the standard and crucial observation: when two ants collide and reverse directions instantaneously, this is indistinguishable (with respect to the set of positions occupied over time) from the ants simply passing through each other (but keeping their original velocities). Equivalently, after a collision the identities of the two ants swap; if we are only concerned with times at which ants fall off the string (not with which labelled ant falls where), we may treat ants as non-interacting — each ant moves straight to an end and falls off without being affected by collisions.

Under this viewpoint, for each ant independently:

- If an ant is at position $x \in [0,1]$ and chooses left, its fall time is x minutes.
- If it chooses right, its fall time is $1-x$ minutes.

Since the initial position x is uniform on $[0,1]$ and direction is independent with probability $1/2$ each, the distribution of the fall time T of a single ant is

$$T = \begin{cases} X & \text{with probability } 1/2, \\ 1-X & \text{with probability } 1/2, \end{cases}$$

where $X \sim \text{Unif}(0,1)$. Both X and $1-X$ are $\text{Unif}(0,1)$, so T is $\text{Unif}(0,1)$. Moreover, fall times of different ants are independent (because positions and directions were independent).

Let $T_1, \ldots, T_{500}$ be the independent fall times (each $\text{Unif}(0,1)$). The time until all ants have fallen off is the maximum

$$T_{\max} = \max_{1 \le i \le 500} T_i.$$

From part (a) (the formula for the maximum of n i.i.d. $\text{Unif}(0,1)$ variables),

$$\mathbb{E}[T_{\max}] = \frac{500}{500+1} = \frac{500}{501}.$$

Numerically,

$$\mathbb{E}[T_{\max}] = \frac{500}{501} \approx 0.99800399 \text{ minutes} \approx 59.88 \text{ seconds}.$$

$$\boxed{\mathbb{E}[\text{time until all ants fall}] = \frac{500}{501} \text{ minutes} \approx 0.9980 \text{ min}.}$$

16. A factory produces a random number of items each day. The number of items produced per day, denoted by N, follows a Poisson distribution with mean $\lambda = 4$. Each batch of items also has a random defect rate P, which varies from day to day due to changes in temperature and humidity. The defect rate P is uniformly distributed on $(0, 0.2)$.

Given $P = p$ and $N = n$, the number of defective items in that day, D, conditionally follows a Binomial distribution: $D \mid P = p, N = n \sim \text{Binomial}(n, p)$. That is,

$$P(D = k \mid P = p, N = n) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, 2, \ldots, n.$$

Find the expected number of defectives per day, $E[D]$.

Solution:

By the Law of Iterated Expectations,

$$E[D] = E[E[D \mid (P, N)]].$$

Given $P = p$ and $N = n$,

$$E[D \mid P = p, N = n] = np.$$

Therefore,

$$E[D] = E[NP].$$

Since N and P are independent,

$$E[D] = E[N]\,E[P].$$

Compute each term:

$$E[N] = \lambda = 4, \qquad E[P] = \frac{0 + 0.2}{2} = 0.1.$$

Hence,

$$E[D] = 4 \times 0.1 = 0.4.$$

Without using the Law of Iterated Expectations, one would need to integrate over the joint distribution of (N, P), involving both the Poisson and Uniform densities. The law simplifies this by allowing us to compute

$$E[D] = E[E[D \mid P, N]] = E[NP],$$

and then factorize the expectation due to independence. This approach avoids complex integration and highlights the power of the Law of Iterated Expectations in hierarchical (multi-stage) random processes.

17. Suppose that a point X_1 is chosen from the uniform distribution on the interval $[0, 1]$, and that after the value $X_1 = x_1$ is observed, a point X_2 is chosen from a uniform distribution on the interval $[x_1, 1]$. Suppose further that additional variables $X_3, X_4, \ldots$ are generated in the same way. In general, for $j = 1, 2, \ldots$, after the value $X_j = x_j$ has been observed, X_{j+1} is chosen from a uniform distribution on the interval $[x_j, 1]$. Find $E[X_n]$.

Solution: We have $X_1 \sim \text{Unif}(0, 1)$, so $E[X_1] = \frac{1}{2}$. For $j \geq 1$, conditioning on $X_j = x$,

$$X_{j+1} \mid X_j = x \sim \text{Unif}(x, 1) \quad \Longrightarrow \quad E[X_{j+1} \mid X_j = x] = \frac{x + 1}{2}.$$

Apply the Law of Iterated Expectations:

$$m_{j+1} \equiv E[X_{j+1}] = E[E[X_{j+1} \mid X_j]] = E\left[\frac{X_j + 1}{2}\right] = \frac{E[X_j] + 1}{2} = \frac{m_j + 1}{2},$$

with $m_1 = \frac{1}{2}$.

Solve the linear recursion. Let $d_j = 1 - m_j$. Then

$$d_{j+1} = 1 - m_{j+1} = 1 - \frac{m_j + 1}{2} = \frac{1 - m_j}{2} = \frac{d_j}{2}.$$

Since $d_1 = 1 - m_1 = 1 - \frac{1}{2} = \frac{1}{2} = 2^{-1}$, it follows that $d_j = 2^{-j}$ and hence

$$E[X_n] = m_n = 1 - d_n = 1 - 2^{-n}.$$

$$\boxed{E[X_n] = 1 - 2^{-n}.}$$

18. Let $X_1, X_2, \ldots, X_n$ be independent random variables such that

$$X_i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad i = 1, 2, \ldots, n.$$

Define

$$S_n = \sum_{i=1}^{n} X_i.$$

1. Derive the moment generating function (MGF) of a single Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ explicitly from its integral definition.
2. Using the law of MGFs and the independence of X_i, find the MGF of S_n and identify its distribution.

Solution: (1) MGF of a single Gaussian random variable.

For $X \sim \mathcal{N}(\mu, \sigma^2)$, the moment generating function is defined as

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx,$$

where

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Thus,

$$M_X(t) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(tx - \frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

We now simplify the exponent by completing the square in x:

$$\begin{aligned} tx - \frac{(x-\mu)^2}{2\sigma^2} &= -\frac{1}{2\sigma^2}\left[(x-\mu)^2 - 2\sigma^2 t(x-\mu)\right] + \mu t \\ &= -\frac{1}{2\sigma^2}\left[(x-\mu-\sigma^2 t)^2 - (\sigma^2 t)^2\right] + \mu t \\ &= -\frac{(x-(\mu+\sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2}. \end{aligned}$$

Substitute this back into the integral:

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-(\mu+\sigma^2 t))^2}{2\sigma^2}\right) dx.$$

The integral term equals 1 because it is the total probability of a normal density with mean $(\mu+\sigma^2 t)$ and variance σ^2. Therefore,

$$\boxed{M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).}$$

—

(2) MGF of the sum of independent normals.

If $X_1, X_2, \ldots, X_n$ are independent, then by the *law of MGFs*,

$$M_{S_n}(t) = M_{X_1+X_2+\cdots+X_n}(t) = \prod_{i=1}^{n} M_{X_i}(t).$$

Using the result from part (1),

$$M_{X_i}(t) = \exp\left(\mu_i t + \frac{1}{2}\sigma_i^2 t^2\right).$$

Hence,

$$\begin{aligned} M_{S_n}(t) &= \prod_{i=1}^{n} \exp\left(\mu_i t + \frac{1}{2}\sigma_i^2 t^2\right) \\ &= \exp\left(\left(\sum_{i=1}^{n} \mu_i\right) t + \frac{1}{2}\left(\sum_{i=1}^{n} \sigma_i^2\right) t^2\right). \end{aligned}$$

This is the MGF of a normal distribution with

$$\text{mean } \sum_{i=1}^{n} \mu_i, \quad \text{variance } \sum_{i=1}^{n} \sigma_i^2.$$

Therefore,

$$\boxed{S_n \sim \mathcal{N}\left(\sum_{i=1}^{n} \mu_i,\ \sum_{i=1}^{n} \sigma_i^2\right)}.$$